ALGEBRA 1 · UNIT 1

Section 1.4 Expressions and Structure

Expressions are more than symbols. Their structure tells a story about quantities, groups, parts, and relationships.

Mathematical idea: Rewriting an expression should preserve its value while making a different feature of the quantity easier to see.

Today we will practice interpreting expression parts, rewriting equivalent expressions, analyzing errors, and choosing forms that reveal meaning.

Learning Target

Students will interpret and rewrite algebraic expressions by identifying structure, meaning, and equivalent forms.

I can statements

- I can identify parts of an expression.
- I can simplify expressions.
- I can write equivalent expressions.
- I can explain what an expression means in context.
- I can use structure to choose an efficient rewrite.

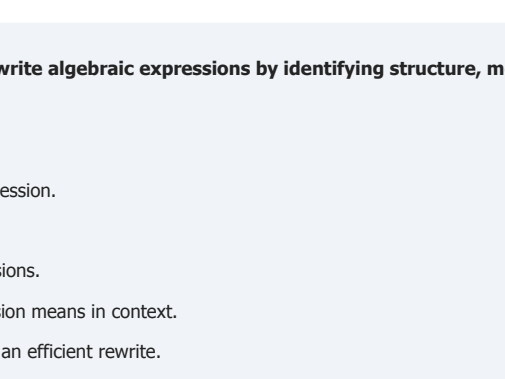
What's To Come

This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: Which student's interpretation could be correct? What is missing from the model? What would need to be true for each expression to be right?

Future Task — Ambiguous Visual Model

A student draws an area model to represent an expression. The height is labeled 3 and the left section's width is labeled x, but the right section's width is unlabeled. Two students disagree about what expression the model represents.


A student's visual model — label unknown
3
x
?
Student A: "The expression is 3(x + ?)"
Student B: "The expression is 3x + ?"
What information is missing?

1. Circle the dimension that is missing from the model.
2. Underline one part that is the same in both students' expressions.
3. Explain why both students could be correct given only the information shown. What specific piece of information would let you decide which expression is right?

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

An expression is a mathematical phrase. It can include numbers, variables, operations, and grouping symbols. Unlike an equation, an expression does not have an equals sign.

Expressions can describe quantities. For example, if a club charges a fee and then charges more for each item, one part of the expression might show the fee and another part might show the repeated cost.

Structure means how the parts of an expression are arranged. The expression $4x + 12$ shows two separate parts being added. The expression $4(x + 3)$ shows a group being multiplied by 4. Sometimes expressions look different but are equivalent, which means they always have the same value for the same input.

Rewriting expressions is useful when a new form reveals something important. One form might show the total as parts. Another form might show groups. A strong algebra student does not just rewrite symbols; they explain what the structure means.

Stop and Discuss

1. The reading says expressions can include numbers, variables, operations, and grouping symbols. According to the reading, what is the one thing an expression does *not* have that an equation does?
2. The reading says equivalent expressions "always have the same value for the same input." Find that sentence and read it aloud. What does the word "always" tell you about what it takes to prove equivalence?
3. When might a grouped form like $4(x + 3)$ be more useful than the separated form $4x + 12$? Give a real-world reason.

Expressions Have Parts

MATH NOTES **Parts of an Expression**

An algebraic expression can be read by looking at its parts and how those parts are connected.

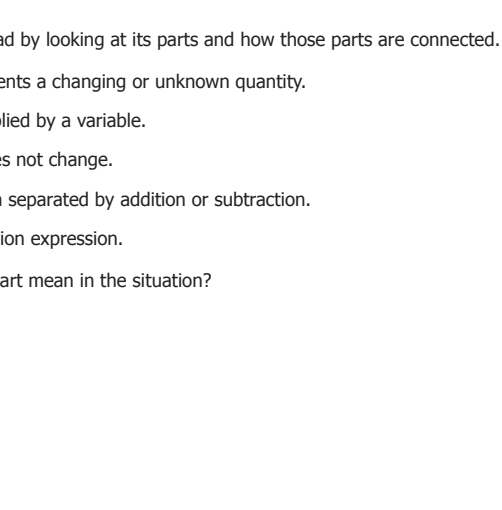
- **Variable:** a letter that represents a changing or unknown quantity.
- **Coefficient:** a number multiplied by a variable.
- **Constant:** a number that does not change.
- **Term:** a part of an expression separated by addition or subtraction.
- **Factor:** a part of a multiplication expression.

Key question: What does each part mean in the situation?

YOUR NOTES

Example 1: Interpret the Structure

A classroom buys 3 identical supply kits. Each kit contains x markers and 4 glue sticks. A student writes the expression $3(x + 4)$.


One possible visual model
3 kits
Kit 1
Kit 2
Kit 3
What do the two sections of each kit represent?

1. Look at the unlabeled sections in the visual model. What two things does each kit contain? Write those two things as a group using parentheses.
2. The model shows 3 identical rows. What does the expression look like when you write 3 groups of your answer from Q1?
3. If each kit has 6 markers (x=6), how many total items are in all 3 kits? Show how the expression gives that answer, not just the arithmetic.
4. Explain why $3(x + 4)$ and $3x + 12$ describe the same total. Which form makes the kits more visible? Which form makes the total items more visible?

You Try It 1: Compare Two Structures

A party planner buys 5 snack bags. Each bag has c crackers and 2 fruit snacks. Two students write expressions for the total number of snacks:

Student A: $5(c + 2)$

Student B: $5c + 10$

1. What part of the situation is easiest to see in Student A's expression?
2. What part of the situation is easiest to see in Student B's expression?
3. Do both expressions represent the same quantity? Use the context as evidence.

MATH NOTES **Equivalent Expressions**

Equivalent expressions are expressions that always have the same value for the same input.

- **Combining like terms** rewrites expressions by joining terms with the same variable part.
- **The distributive property** rewrites a product over a sum or difference.
- **Substitution** can check whether two expressions give the same value for a chosen input, but one input alone does not prove equivalence forever.

Useful test: Does the new expression keep the same quantity, or did something change?

YOUR NOTES

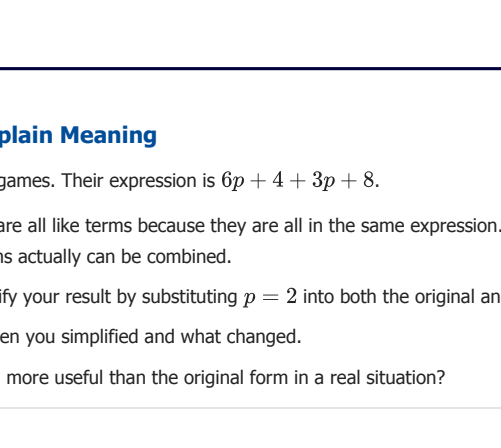
Example 2: Simplify and Explain Meaning

A student is tracking points from two games. Their expression is $6p + 4 + 3p + 8$.

1. A student says $6p$, $3p$, 4, and 8 are all like terms because they are all in the same expression. Explain what is wrong with that reasoning and identify which terms actually can be combined.
2. Simplify the expression. Then verify your result by substituting $p = 2$ into both the original and simplified forms.
3. Explain what stayed the same when you simplified and what changed.
4. Why would the simplified form be more useful than the original form in a real situation?

You Try It 2: Verify Equivalent Forms

A student claims $5(n + 2)$ and $5n + 10$ are equivalent.


Visual model for 5(n + 2)
5
?
?
Label each section with what it represents.

1. The visual model shows a rectangle divided into two sections. Label each section with what it represents in the expression $5(n + 2)$. Explain how the model shows the total area equals $5n + 10$.
2. Check both expressions when $n = 3$. What do you notice? Then check again for $n = 0$. Does the result change?
3. Does checking two values prove equivalence forever, or does it only provide supporting evidence? Explain what would be required to prove equivalence completely.

Stop and Discuss

A student says, "If two expressions have the same numbers and variables, they must be equivalent." What is wrong with that reasoning?

MATH NOTES **Analyzing Expression Errors**

When checking student work, look for the moment when the structure changed incorrectly.

1. Identify the original terms, groups, and operations.
2. Ask which terms are actually like terms.
3. Check whether distribution reached every term inside the group.
4. Use substitution or context to test whether the value stayed the same.

ORIGINAL STRUCTURE

WHERE IT CHANGED

Example 3: Analyze the Error

A student simplifies $3x + 4 + 2x$ as $9x$.

Student work: $3x + 4 + 2x = 9x$ because $3 + 4 + 2 = 9$.

1. What did the student combine incorrectly?
2. Why is the constant 4 not like $3x$ or $2x$?
3. Write the correct simplified expression.
4. Write one sentence that would help the student avoid this mistake next time.

You Try It 3: Critique the Reasoning

A student rewrites $4(x + 6)$ as $4x + 6$.

Student explanation: "I multiplied the x by 4, and the 6 was already there."

1. The student says "the 6 was already there." What does that phrase reveal about their misconception — what did they think the parentheses meant, and what do parentheses actually mean in this context?
2. Use substitution with $x = 2$ in both $4(x + 6)$ and $4x + 6$ to show they are not equivalent. Explain what the numbers tell you.
3. Write the correct equivalent expression. Then write one sentence of advice that addresses the exact misconception in the student's explanation.

MATH NOTES **Choosing a Useful Form**

Equivalent expressions can be useful in different ways.

- A **separated form**, such as $4x + 12$, can show parts being added.
- A **factored or grouped form**, such as $4(x + 3)$, can show equal groups or a shared factor.
- A **context form** uses words to explain what each part means.

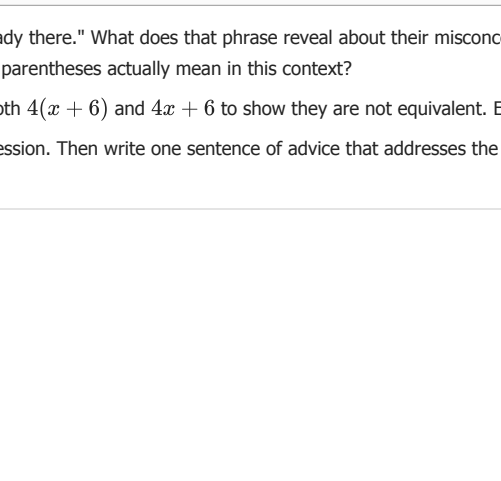
Decision question: Which form makes the important idea easiest to see?

YOUR NOTES

Example 4: Compare Equivalent Forms

A school club buys supplies. Two equivalent expressions are shown:

$$4x + 12 \qquad \text{and} \qquad 4(x + 3)$$


Two useful forms for the same quantity
4x
12
One possible visual model

1. Use distribution to show step-by-step why $4(x + 3)$ is equivalent to $4x + 12$. Show each multiplication explicitly.
2. A club treasurer wants to quickly see how many items are in each equal group. Which form is more useful and why? A budget analyst wants to see the total add-on cost of 12 units. Which form is more useful and why?
3. If you wanted to rewrite $4x + 12$ in a factored form that shows a factor of 4, what would you write? What if the factor you pulled out were 2 instead — what would the new factored form be?
4. Write a possible context where the factored form reveals something important that the expanded form hides.

You Try It 4: Create an Equivalent Representation

A gym rents equipment for $7 per person and charges one group fee of $14.

1. Write an expression for the total cost for p people.
2. Create a different equivalent expression or visual model for the same situation.
3. Explain what each part of your expression or model represents.
4. Which form would be most useful if someone wanted to see the group fee quickly? Explain.

Summary

Expressions describe quantities using variables, constants, terms, factors, operations, and grouping symbols. The structure of an expression shows how the parts of a quantity are connected.

Equivalent expressions may look different but keep the same value for the same input. Rewriting can make an expression easier to simplify, easier to connect to a context, or easier to compare with another representation.

Strong algebraic reasoning means checking whether the meaning stayed the same, not just moving symbols around.

Common Mistakes

- **Combining unlike terms.** Why it is tempting: all the symbols are in the same expression. Check instead: only combine terms with the same variable part.
- **Distributing to only one term.** Why it is tempting: students often multiply the first visible term and stop. Check instead: the outside factor must multiply every term inside the group.
- **Assuming same numbers means equivalent.** Why it is tempting: the expressions look related. Check instead: test the operations and structure, not just the numbers.
- **Ignoring context meaning.** Why it is tempting: symbolic work can feel separate from the story. Check instead: ask what each term, factor, and group represents.
- **Thinking simplified form is always best.** Why it is tempting: simplified expressions look shorter. Check instead: choose the form that reveals the information you need.

Optional Future Task Reflection

Look back at the future task. Do not solve it yet. Highlight one part that makes more sense now than it did at the beginning. Circle one part that still feels unfamiliar. Write one sentence about what representation you would look at first.